

Q.No:7

DS	Session	Searching	Question Information	Level 1	Challenge 7
Question Description: Suresh have "N" rectangles. A rectangle is Silver if the ratio of its sides is in between $[1.6, 1.7]$, both inclusive. Your task is to find the number of silver rectangles. Constraints: $1 \leq N \leq 10^5$ $1 \leq W,$ $H \leq 10^9$ Input Format: First line: Integer "N" denoting the number of rectangles Each of the "N" following lines: Two integers W, H denoting the width and height of a rectangle Output Format: Print the output in a single line contains find the number of Silver rectangles. Sample Input: 5 10 1 165 100					

180 100

170 100

160 100

Sample Output:

3

Explanation:

There are three Silver rectangles: (165, 100), (170, 100), (160, 100).

Logical Test Cases

Test Case 1
INPUT (STDIN)
4
185 100
180 100
170 100
160 100
EXPECTED OUTPUT
2

Test Case 2
INPUT (STDIN)
6
185 100
180 100
170 100
160 100
165 90
164 95
EXPECTED OUTPUT
2

Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
int N;
long long width, height;
int count = 0;

scanf("%d", &N);

for (int i = 0; i < N; i++)
{
scanf("%lld %lld", &width, &height);

if (width >= height)
{
if (width * 10 >= height * 16 &&
width * 10 <= height * 17)
{
count++;
}
}
else
{
if (height * 10 >= width * 16 &&
height * 10 <= width * 17)
{
count++;
}
}
}

printf("%d", count);

return 0;
}
```

Output:

Test_case 1

```
4
185 100
180 100
170 100
160 100
2%
```

Test_case 2:

```
6
185 100
180 100
170 100
160 100
165 90
164 95
2%
```